

Yumna Islam

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EDUCATION

Ahsanullah University of Science and Technology

Dhaka, Bangladesh

B. Sc. in Computer Science and Engineering

Fall 2018–Spring 2022

CGPA: 3.809 on a scale of 4.00 (7th in the graduating class of 102 students)

RESEARCH INTERESTS

Computer Vision, Machine Learning, Deep Learning, Embedded Systems, Robotics, Human Robot Interaction

PUBLICATIONS

- **Yumna Islam**, Farzad Azad, C.Z. Md Ruslan, C. Aye Mong Marma, and Kazi A Kalpoma. **A Lane Detection Framework for Automated Vehicle Using Computer Vision**. Published in *International Conference on Power Engineering and Intelligent Systems (PEIS 2023)*, Lecture Notes in Electrical Engineering, vol 1098, Springer, Singapore. (2024)
- Farhana Azad, **Yumna Islam**, C.Z. Md Ruslan, C. Aye Mong Marma, and Kazi A Kalpoma. **Efficient Lane Detection and Keeping for Autonomous Vehicles in Real-World Scenarios**. Published in *26th International Conference on Computer and Information Technology (ICCIT 2023)*, Cox's Bazar, Bangladesh. *IEEE*, pp. 1–6. (2023)

RESEARCH EXPERIENCE

Undergraduate Thesis

2022 – 2023

Using Automotive ECU and Computer Vision Based Image Processing for Vehicle Lane Keeping (Supervisor - Prof. Dr. Kazi A Kalpoma)

AUST, Bangladesh

- Implemented an image processing pipeline for lane detection in 5 different lighting conditions (iROADS dataset), achieving an average detection rate of 95.68%.
- Simulated lane keeping with warning feature via AURIX microcontroller

Natural Language Processing Course Project

2022

Bengali Literature Author Identification Using Pre-trained Models

AUST, Bangladesh

- Fine-tuned 4 different pre-trained transformer models (BERT) for detecting authorship, mainly from Bengali corpus (BAAD 16) and analyzed how oversampling and augmentation can improve overall accuracy

Pattern Recognition Course Project

2022

Intrusion Detection Using Multiple Machine Learning Models

AUST, Bangladesh

- Investigated multiple machine learning model performance for intrusion detection (dataset NSL-KDD), focusing on feature engineering

TEACHING EXPERIENCE

Private Tutor

2019 – 2022

Dhaka, Bangladesh

Tutored 10+ high school students (grade 11 & 12) on Physics, Mathematics and ICT topics - Electromagnetism, Calculus, Algebra, Computer Fundamentals and Basic Programming. Planned lessons for easier applications of elementary programming and graded maths and physics papers.

PROFESSIONAL EXPERIENCE

Software Engineer (Test)

Feb 2025 - Present

Meghna Bank Ltd, Dhaka, Bangladesh

- Identified software defects via UAT, API testing and resolved them by collaborating with PMs, developers, and business teams across projects, including in-house utility bill collection solution, corporate payment integration, VISA remittance, and agent banking. Mentored 1 intern on software development process and product life cycle

QA Associate (Solution Design & QA)

Oct 2023 - Aug 2024

Nagad Ltd, Dhaka, Bangladesh

- Documented functional and UI issues of mobile & web portal in JIRA and performed exploratory, system, regression testing to ensure bug free product for one of the biggest finetch in Bangladesh

Technology Intern (Software QA)

Jul 2023 - Sept 2023

Nagad, Dhaka, Bangladesh

- Assisted in manual testing and documentation of user interface inconsistencies and obtained holistic view on agile software development and testing cycle

TECHNICAL SKILLS

Languages: Python, C, C++, JAVA, HTML5, PHP, JavaScript, 80x86 Assembly

Framework: CSS, WebGL

Database: MySQL, MS SQL

Design Tools: NetBeans, Microsoft Visual Studio, PyCharm, Jupyter Notebook, Google Colab

Deep Learning Tools: Tensorflow, Pytorch

Others: PL/SQL, L^AT_EX

ACADEMIC PROJECTS

- SIMS- Student Internship Management Systems (Tools: Java Swing, MS SQL & PL/SQL)
- Fashion & Lifestyle Blogsite- ALOHA (Tools: HTML5, CSS, Javascript, PHP, MySQL with an ASP.NET version)
- Cooling Load Prediction of a Building (Tools: Python, Matplotlib, Google Colab)
- Covid19 Vaccine Management System (Tools: Android Studio, Java, MS SQL)
- Blind Stick (Tools: Arduino UNO, C++)

AWARDS & ACHIEVEMENTS

- "Dean's List of Honor" (based on B.Sc. result)
- 50% Tuition Waiver for 2 semesters based on semester final results during B.Sc. (Awarded to top 5% students in a specific department)
- Certificates for completing the courses "Supervised Machine Learning: Regression and Classification" & "Advanced Learning Algorithms" in Coursera